Journal of Cleaner Production 360 (2022) 132239

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Evaluation of deep eutectic solvent pretreatment towards efficacy of enzymatic saccharification using multivariate analysis techniques

Salim Massayev, Kiat Moon Lee *

Department of Chemical & Petroleum Engineering, Faculty of Engineering, Technology and Built Environment, UCSI University, 56000, Kuala Lumpur, Malaysia

ARTICLE INFO

Handling Editor: M.T. Moreira

Keywords:
Lignocellulosic biomass
Deep eutectic solvent
Pretreatment
Multivariate analysis
Principal component analysis
Partial least square

ABSTRACT

In this paper, deep eutectic solvent (DES) pretreatment of lignocellulosic biomass was analysed using principal component analysis (PCA) and partial least square (PLS) methods to determine correlations between the pretreatment process variables and assess their significance level. The analysis was conducted on the dataset containing 103 experimental observations and 32 variables. Both PCA and selected PLS model (M4) had substantial prediction power and prediction relevance, as exhibited by their good cumulative explained variance (R2cum, 0.787) and cumulative predicted variance (Q2cum, 0.524) for six principal components of PCA model, and explained variance of independent variable (R2Xcum, 0.781), explained variance of dependent variable (R2Ycum, 0.803) and cumulative predicted variance (Q2cum, 0.776) for the best PLS model. The results of PCA and PLS analyses revealed that the most significant variables for DES pretreatment were severity factor, temperature, type of antisolvent, particle size, agitation intensity and type of hydrogen bond donor (HBD) of the DES. The use of carboxylic acid and hydroxy acid HBD was associated with higher glucose yield and shorter duration of saccharification. The models also showed that enzymatic saccharification was more efficient when organic solvent or inorganic salt solution was used as antisolvent for lignocellulose regeneration. Usage of water as antisolvent was associated with reduced glucose yield during saccharification. Another notable outcome of this work was that intensive stirring during pretreatment had negative effect on glucose yield due to the shear thickening properties of DESs. This work provided a systematic analysis of DES pretreatment process of lignocellulosic biomass using PCA and PLS methods that allowed better understanding of the factors affecting the pretreatment process.

1. Introduction

The global trends to waste minimization and reduction of fossil fuel usage stipulated great research interest to alternative energy sources. Lignocellulose is one of the most abundant renewable sources of carbohydrates and therefore it is viewed as a primary candidate for producing biofuels. The major challenge in lignocellulose bioconversion is the high recalcitrance of raw lignocellulosic biomass, that is attributed to the presence of crystalline cellulose regions in the lignocellulose structure (Lee et al., 2021), high resistance of lignin to chemical attacks (Lauberte et al., 2021), as well as the complex polymeric structure of lignocellulose (Lobato-Peralta et al., 2021). Generally, satisfactory efficacy of lignocellulose bioconversion can be achieved only with the aid of a specific pretreatment that could efficiently disrupt lignin-cellulose bonds and reduce the overall crystallinity of cellulose. Conventional pretreatment methods like acid hydrolysis, pyrolysis, alkaline treatment, organic solvent treatment, ozonation, steam explosion, carbon dioxide explosion, and ammonia fiber explosion involve harsh treatment conditions resulting into partial loss of sugars due to degradation (Mahmood et al., 2016). Non-conventional pretreatment methods such as ultrasonic (Lee et al., 2020), ionic liquid (IL) (Lee et al., 2015), and deep eutectic solvent (DES) (Thi and Lee, 2019) are conducted at mild conditions and therefore can boast significantly lesser sugar loss. Within this paper, DES pretreatment was considered. DES pretreatment is a promising method for lignocellulose pretreatment that features superior biocompatibility, low environmental impact, and versatility (Abbott et al., 2004).

DES is a mixture of at least two components one of which is hydrogen bonding acceptor (HBA) and another is hydrogen bonding donor (HBD) whose melting point is significantly lower than that of the individual components and generally not exceeding 100 °C (Abbott et al., 2004).

* Corresponding author. Department of Chemical & Petroleum Engineering, Faculty of Engineering, Technology and Built Environment, UCSI University (Kuala Lumpur Campus), UCSI Heights, 1, Jalan Puncak Menara Gading, Taman Connaught, 56000 Cheras, Kuala Lumpur, Malaysia.
E-mail address: leekm@ucsiuniversity.edu.my (K.M. Lee).

https://doi.org/10.1016/j.jclepro.2022.132239
Received 20 July 2021; Received in revised form 4 May 2022; Accepted 12 May 2022
Available online 16 May 2022
0959-6526/© 2022 Elsevier Ltd. All rights reserved.

List of abbreviations

α –	DES hydrogen bond acidity in Kamlet-Taft parameters
β –	DES hydrogen bond basicity in Kamlet-Taft parameters
π –	DES dipole polarization
μ –	DES viscosity at 25 °C
AS –	antisolvent
BP –	boiling point
CN –	carbon number of HBD
D –	dipole moment
DES –	deep eutectic solvent
HBD –	hydrogen bond donor
IL –	ionic liquid
MP –	melting point
MW –	molecular weight
PCA –	principal component analysis
PLS –	partial least squares
P-time –	pretreatment duration
Q2cum –	cumulative predicted variance
R2cum –	cumulative explained variance
R2Xcum –	explained variance for X variables
R2Ycum –	explained variance for Y variables
SF –	severity factor
S/L –	solid-to-liquid ratio
S-time –	saccharification duration
t –	temperature

There are four groups of DESs depending on their HBA and HBD. Type I DESs are composed of organic salts, most frequently a quaternary ammonium salt, and non-hydrated metal halides such as zinc chloride ($ZnCl_2$), iron (III) chloride ($FeCl_3$), copper (II) chloride ($CuCl_2$), lithium chloride (LiCl) and silver chloride (AgCl). Hydrated metal halides are introduced in type II DES. In type III DESs, HBD is represented by a polyalcohol, organic acid, or amide, while HBA remains same the same as type I and type II DESs, *i.e.* organic salt. While type IV DESs are formed by eutectic mixtures consist of inorganic transition metals with HBD such as urea, ethylene glycol, 1,6-hexanedoil and acetamide. Type III DESs are the most studied type due to easily synthesis, non-reactivity with water, wide range solubility of transition metal, biodegradable and economically feasible (Loow et al., 2017). DES is widely recognized as a green solvent due to its low toxicity, volatility, flammability, and high recyclability (Zdanowicz et al., 2018).

Multiple researchers have reported favorable changes in the structure and composition of lignocellulosic biomass after DES treatment, such as selective removal of hemicellulose and lignin, reduction of cellulose crystallinity (Sai and Lee, 2019), and increases in biomass particles surface area (Lee et al., 2021), which lead to enhanced enzymatic saccharification efficiency and productivity. Most of these works were focused on small number of process variables, such as pretreatment temperature, pretreatment duration, biomass-to-solvent ratio, biomass particle size, and Kamlet-Taft parameters of DES. Generally, the authors altered one parameter of pretreatment process at a time and observed how it would affect pretreatment efficiency. Such approach neglects correlations between process variables other than the selected few. Moreover, it is relatively rare for several independent researchers to have identical basis, which makes it troublesome to compare results of current experiment with those conducted by other researchers.

One of the options to resolve these issues is to employ multivariate data analysis that can simultaneously analyze the effect of multiple variables on the pretreatment process efficacy and compare results of large number of research data while considering differences in experiments bases (Li et al., 2017). Principal component analysis (PCA) and partial least square (PLS) are two of the most used techniques for multivariate data analysis. PCA is commonly used as a first-look tool to identify relationships between variables that can be further investigated in detail with other techniques (Mahmoudi et al., 2019). It employs latent variables to reduce the dimensionality of the initial dataset and determine the correlation between independent variables. On the other hand, PLS is utilized to evaluate the significance of the relationships between selected dependent and independent variables (Hajilary et al., 2018). It develops a linear regression model by projecting the predicted dependent variable matrix and the observed independent variable matrix to a latent variable space. Therefore, both PCA and PLS can be used to estimate the contribution of different variables to the pretreatment efficacy.

In this study, multivariate analysis of DES pretreatment of various types of lignocellulosic biomass was conducted to fill this knowledge gap. To the best knowledge of the authors, only one multivariate analysis study has been conducted on DES pretreatment of lignocellulose by Xu et al. (2020). However, some of the important pretreatment parameters such as type of antisolvent, stirring rate, and saccharification duration were not considered. In this work, all the important variables in DES pretreatment including pretreatment conditions, type of antisolvent, biomass composition, type of DES and its physicochemical properties were investigated. The main objectives of this work were to investigate the relationships between various parameters of green solvent pretreatment of lignocellulosic biomass using PCA, to study the significance of various parameters of green solvent pretreatment using PLS and to evaluate the quality of the developed PCA and PLS models.

2. Methodology

2.1. Data collection

Dataset on lignocellulosic biomass pretreatment with various DESs was obtained by tabulation of experiment conditions and results published by other researchers in the period from 2004 to present. Different species of lignocellulosic biomass were included in this study. All the data were extracted from various credible online journals and scientific databases such as ScienceDirect, SpringerLink, and ACS Publications.

2.2. Multivariate analysis

In this work, PCA and PLS analysis of the dataset containing 103 observations and 27 important variables of DES pretreatment was performed with the use of SIMCA 14.1.0.2047 (Soft Independent Modelling of Class Analogy, Umetrics). Variables considered in PCA and PLS were: pretreatment conditions (temperature, pretreatment duration, severity factor, solid-to-liquid loading, biomass particle size and agitation intensity), type of antisolvent (pure water, organic and inorganic solvent (*i.e.* water with additives)), biomass composition (cellulose, hemicellulose, and lignin content), general characteristics and type of HBD (number of hydroxyl groups, carboxyl groups, amino groups, and class of chemical compound: polyol, hydroxyl acid, carboxylic acid, or amide), physicochemical properties of DES and HBD (HBD molecular weight, HBD carbon number, HBD melting point, HBD boiling point, HBD pKa, viscosity, DES melting point, Kamlet-Taft hydrogen bond acidity, Kamlet-Taft hydrogen bond basicity, and Kamlet-Taft dipolarity-polarizability), and enzymatic hydrolysis efficacy metrics (saccharification duration and glucose yield). List of variables investigated with their corresponding abbreviations used in the models can be found in Appendix A.

2.3. Data preprocessing

Both PCA and PLS models cannot work with categorical variables, therefore all categorical variables were replaced with dummy variables. For example, non-binary categorical variable "type of HBD" which can be either polyol, hydroxy acid, amide, or carboxylic acid was replaced with four binary numeric variables, where "1" indicated that certain HBD type was used and "0" indicated the opposite. The variable type for all the investigated variables is presented in Table A1. To maintain consistency, glucose yields that were expressed in terms of weigh of glucose per weigh of biomass were converted to weigh of glucose per weigh of cellulose by dividing glucose yield expressed on the biomass weight basis by cellulose content in the pretreated biomass. Biomass particle sizes that were presented in the form of fraction remaining between two mesh sizes after screening were expressed as the arithmetic average of the maximum and minimum particle size in that fraction.

3. Results and discussion

In this work, the effect of different DES pretreatment conditions on the efficiency of lignocellulose enzymatic hydrolysis was investigated using multivariate data analysis with PCA and PLS methods. Variables considered in PCA and PLS are shown in Table 1.

3.1. Principal component analysis of DES pretreatment of lignocellulosic biomass

PCA works through dimensional reduction of large data set by transforming a large set of variables into a smaller data set with most of the information retains. With a smaller data set, data analysis becomes easier and faster. The first principal component (p[1]) was calculated such that it accounts for the greatest possible variance in the dataset. Every successive principal component account for smaller variance than the previous one. In PCA model M1, six principal components accounted for 78.7% of the total variability of the dataset. From Table 2, cumulative explained variance (R2cum) and cumulative predicted variance (Q2cum) for six principal components were 0.787 and 0.524, respectively. The value of R2cum represents the proportion of variable for all the original variables. The closer its value to 1, the better the analysis results of the PCA model. As rule of thumb, models with R2cum above 0.75 are said to have substantial predictive power (Eriksson et al., 2013). Q2cum is the measure of the model's prediction relevance. Models with the value of Q2cum above 0.5 and the difference R2cum and Q2cum not exceeding 0.2 to 0.3 are considered to have good predictive relevance (Eriksson et al., 2013). Therefore, the PCA model M1 was sufficiently reliable to explain the DES pretreatment process.

To determine relations between different variables, three loading scatter plots with the highest combined explained variance for a pair of principal components were analyzed. On the loading scatter plot, variables located close to each other within one quadrant are positively correlated, while negatively correlated variables appear in the opposite quadrants relative to the origin. The distance from the point of origin to a certain variable reveals the degree of influence of the variable on the model. The further away a variable located from the origin the more influence it has on the overall model, while the variables located close to the origin point have a low contribution to the model.

Fig. 1 displays the loading scatter plots for three principal components that accounted for the highest variability. p[1] and p[2], p[1] and p[3], p[1] and p[4], collectively accounted for 43.8%, 42.5%, and 40.9% of the total variability. p[1] was strongly affected by DES melting point, pKa of HBD, and polyol type HBD. p[2] was strongly dominated by Kamlet-Taft hydrogen bond basicity and dipolarity-polarizability. From this, it was concluded that the physicochemical properties of DES had a significant effect on the pretreatment process in the analyzed model. The finding is supported by Scelci et al. (2021), where acidity (pKa), hydrogen bond capacity and polarity are the most influencing properties associated with DESs in biomass pretreatment. p[3] was dominated by HBD properties such as amide HBD and number of amino groups, carbon number of HBD, HBD molecular weight, and HBD boiling point. p[4] was most affected by saccharification duration, HBD boiling point, and HBD carbon number.

Based on the observed scatter plots, pretreatment duration and severity factor were positively related to glucose yield; however, temperature was weakly correlated with glucose yield. The effect of pretreatment duration on glucose yield reported in this work is similar to the conclusion reported by other researchers. For instance, Liu et al. (2019) and Xu et al. (2018) reported that increased in treatment duration led to increase of glucose yield. Similarly, the severity factor was also positively correlated with glucose yield, which coincided with the conclusion of multiple publications (Lloyd and Wyman, 2005; Nguyen et al., 2000; Scelsi et al., 2021; Schell et al., 2003). Scelsi et al. (2021) also reported that severity is one of the most crucial impact on pretreatment performance. Though, it was reported that increase of pretreatment temperature can significantly increase glucose yield due to more efficient disruption of lignin links at elevated temperature

Table 1
List of variables used in PCA and PLS models for DES pretreatment.

	PCA M1	PLS M2	PLS M3	PLS M4	PLS M5
t	X	X	X	X	X
P-time	X	X	X	X	X
SF	X	X	X	X	X
S/L	X	X	X	X	X
Size	X	X	X	X	X
Agitation	X	X	X	X	X
Glucose	X	Y	Y	Y	Y
Water-AS	X	0	X	X	X
Organic-AS	X	0	X	X	X
Inorganic-AS	X	0	X	X	X
S-time	X	Y	Y	Y	Y
Cellulose	X	0	0	X	X
Hemicellulose	X	0	0	X	X
Lignin	X	0	0	X	X
HBD MW	X	0	0	0	X
CN	X	0	0	0	X
Hydroxyl-G	X	0	0	0	X
Carboxyl-G	X	0	0	0	X
Amino-G	X	0	0	0	X
HydroxyAcid	X	0	0	0	X
Polyol	X	0	0	0	X
Amide	X	0	0	0	X
CarboxilicAcid	X	0	0	0	X
D	X	0	0	0	X
HBD MP	X	0	0	0	X
HBD BP	X	0	0	0	X
pKa	X	0	0	0	X
μ	X	0	0	0	X
DES MP	X	0	0	0	X
α	X	0	0	0	X
β	X	0	0	0	X
π	X	0	0	0	X

X – this variable was an independent variable in the selected PCA/PLS model (M).
Y – this variable was a dependent variable in the selected PCA/PLS model (M).
0 – this variable was excluded from the selected PCA/PLS model (M).

Table 2
Cumulative explained and predicted variance for principal components.

Component	R2cum	Q2cum
1	0.310	0.254
2	0.438	0.316
3	0.551	0.345
4	0.651	0.396
5	0.728	0.478
6	0.787	0.524

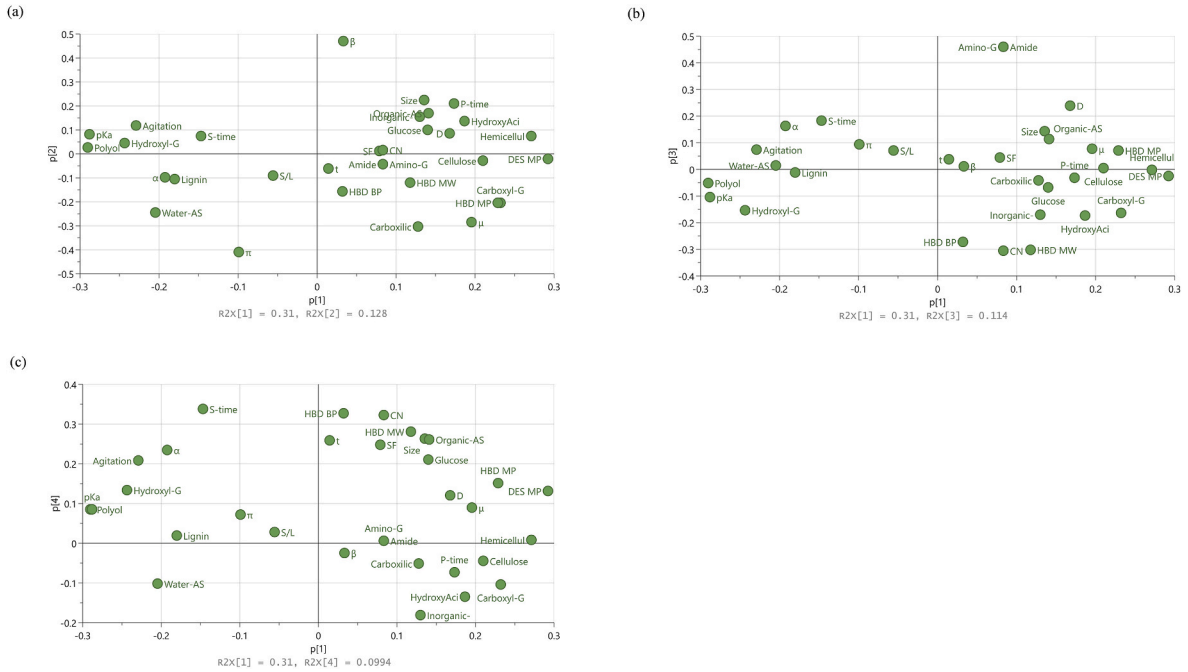


Fig. 1. PCA loading scatter plot for DES pretreatment of lignocellulosic biomass: (a) plane p[1] and p[2]; (b) plane p[1] and p[3]; and (c) plane p[1] and p[4].

(Procentese et al., 2018), pretreatment at too high temperature can also lead to biomass and DES degradation (Sai and Lee, 2019) and clumping of biomass (Lee et al., 2021). which eventually decreases the pretreatment performance. PCA analysis is based on linear transformation of original data, therefore it may not be suitable for temperature effect study on glucose yield. Since severity factor is calculated as a logarithmic function of pretreatment duration and temperature; therefore, according to the results of PCA model analysis, high severity of treatment with prolonged pretreatment duration and reduced temperature should be used to maximize glucose yield.

Different types of antisolvents had varying effect on glucose yield. Organic and inorganic solution antisolvents had a strong positive correlation with glucose yield, while water antisolvent was correlated to glucose yield in an inverse manner. In other words, organic and inorganic solution based antisolvents were more beneficial to glucose yield rather than deionized water. As for the composition of lignocellulose, glucose yield was inversely related to lignin content. Type of HBD also played a significant role in DES pretreatment efficiency. Hydroxy acid and carboxylic acid DES had a strong positive effect on glucose yield. Polyol based HBD and number of hydroxyl groups were positively correlated with saccharification duration. On the other hand, carboxylic acid and amide based HBD, as well as the number of their characteristic functional groups negatively correlated to saccharification duration. In other words, polyol based HBD were associated with lower lignocellulosic biomass pretreatment efficacy. This could be caused by partial condensation of chemical structures of biomass in polyols (Al Ameri, 2017). As for physiochemical properties of DES, Kamlet-Taft hydrogen bond basicity had a weak positive influence on glucose yield, while Kamlet-Taft hydrogen bond acidity and dipolarity-polarizability had negative correlation. On the other hand, hydrogen bond acidity and dipolarity-polarizability had positive correlation with saccharification duration. Besides, saccharification duration also had a weak positive correlation with pKa of HBD and negative relation with viscosity, HBD melting point, and DES melting point.

According to Doherty et al. (2010), pretreatment of lignocellulosic biomass with IL possessing high hydrogen bond basicity resulted in greater glucose and xylose yield. Similar finding were obtained for DES pretreatment (Lee et al., 2021; Ren et al., 2016). These correlates well with the finding of present work. However, correlations of glucose yield with hydrogen bond acidity contradicted the results of the study conducted by Xia et al. (2018). Xia et al. (2018) reported that DESs with higher hydrogen bond acidity shown better efficiency in biomass delignification than DESs with lower hydrogen bond acidity. Thus, hydrogen bond acidity is expected to be positively correlated with glucose yield, while the opposite was shown by the PCA model. A possible cause of this contradiction is the large difference between Kamlet-Taft parameters used by Xia et al. (2018) and in the present study. As shown in Table 3, the difference between the Kamlet-Taft parameter values given by Florindo et al. (2018) and Xia et al. (2018) was quite large. Due to scarcity of publications on Kamlet-Taft parameters of DESs, it is not possible to determine which researcher provided accurate values, therefore values proposed by Florindo et al. (2018) were used in this study since they also provided Kamlet-Taft parameters for several other DESs.

Positive correlation of particle size with glucose yield was obtained in the PCA model. The positive correlation of particle size with glucose yield agreed with the work conducted by Liu et al. (2013). Nevertheless, different findings were reported by other researchers on the effect of particle size on biomass pretreatment. For instance, Semerci and Ersan (2021) reported that moderate particle size of 0.15–1.18 mm was most effective in enhancing pretreatment performance, compared to smaller or larger particle size of <0.15 mm and >1.18 mm. In another work conducted by Wang et al. (2020), better pretreatment performance was reported with smaller biomass size. These suggested that the increase in surface area with the reduction of particle size may not be the sole reason in influencing the pretreatment performance, but other factors such as mass transfer efficiency and changes in viscosity with reduced particle size should be considered. The combined effect may result in non-linear correlation between particle size and glucose yield. As a result, PCA analysis may not be suitable since it is based on linear

Table 3
Kamlet-Taft parameters obtained from different references.

Deep eutectic solvent	α	β	π	Reference
Choline chloride:glycerol (1:2)	1.49	0.52	1.11	Florindo et al. (2018)
Choline chloride:glycerol (1:2)	0.32	0.9	0.82	Degam (2017)
Choline chloride:glycerol (1:2)	1.18	0.77	0.48	Xia et al. (2018)

approximation, which further justify the low R2cum and Q2cum results in later discussion. More research on effect of particle size on pretreatment performance should therefore be performed.

Positive correlation of agitation intensity with the saccharification duration was obtained in the PCA model. It is generally believed that the stirring rate has a positive effect on pretreatment efficacy due to the improvement of mass transfer and reduction of apparent viscosity. However, in the case of non-Newtonian fluids, an increase of shear rate can result in both an increase of apparent viscosity, which is known as a shear thickening, or non-linear decrease of viscosity, which is known as shear thinning. According to Elhamarnah et al. (2019), the majority of DESs are non-Newtonian fluids exhibiting shear thinning behavior. For example, choline chloride:glucose (1:1), choline chloride:citric acid (1:1), choline chloride:glycerol (1:2), and choline chloride:urea (1:2) exhibit decrease of viscosity with increasing shear rate at room temperature. Only one DES was reported to exhibit shear thickening behavior in that study, which is choline chloride:lactic acid (1:1) at 85 °C. Moreover, the presence of suspended solid particles could change the liquid behavior to non-Newtonian liquids, and in most cases resulted in shear thickening (Brown and Jaeger, 2014). Therefore, it can be hypothesized that the addition of finely milled lignocellulosic biomass to DES might produce a mixture that exhibits shear thickening properties. The viscosity of such fluid would increase at a high stirring rate, which in turn would negatively affect pretreatment efficacy. Further research on rheological behavior of DES with the presence of lignocellulosic biomass should be performed.

Fig. 2 displays R2cum and Q2cum for each variable considered in all principal components of the PCA model. The green bars represent R2cum that describes how well a variable can be explained and the blue bars represent Q2cum that can tell how well a variable can be predicted. In other words, the R2cum is the measure of the contribution of a certain variable to the predictive power of the overall model, while Q2cum is a measure of the variable bias or dependence degree on specific data used for model training. A good model is expected to have high predictive power that does not change with inclusion or exclusion of certain observations from the original dataset (Eriksson et al., 2013). Majority of the variables had high explained variance indicating that they were significant to the model. Solid–to-liquid ratio, cellulose, and lignin content had R2cum slightly below 0.5, which means they had a small contribution to the model comparing to other variables. R2cum was sufficiently high only for half of the variables. Q2cum for particle size, organic antisolvent, inorganic antisolvent, saccharification duration, cellulose content, lignin content, dipole moment, HBD carbon number, hydroxy acid, carboxylic acid, HBD boiling point, and hydrogen bond acidity were below 0.5, which indicated that these variables could be biased. One variable that stood out among all other was solid-to-liquid ratio; it was the only variable with negative Q2cum, which meant it had no predictive relevance.

One of the possible causes of low R2cum and Q2cum is experimental data bias. Since the training dataset for this analysis was obtained by the tabulation of conditions and results of experiments performed by other researchers, the obtained data might be biased due to the unreported differences in the treatment and measurement procedures. For example, values of hydrogen bond acidity of choline chloride:glycerol with a molar ratio of 1:2 measured by different researchers were 1.49 (Florindo et al., 2018), 0.32 (Degam, 2017), and 1.18 (Xia et al., 2018), as shown in Table 3. The second reason for low explained and predicted variance is the non-linearity of relations between some of the variables. PCA model calculation is based on the linear approximation of data (Eriksson et al., 2013). If two variables are related non-linearly, the linear approximation will lead to the loss of some portion of data resulting in a less accuracy of the model.

3.2. Partial least square analysis of DES pretreatment of lignocellulosic biomass

PLS is a mathematical optimization technology that determines the best fit function by minimizing the square error. PLS input is a dataset with assigned dependency relations among the variables. The major motivation to use PLS is to verify hypothesized relations between independent and dependent variables. In this study, four different PLS models were studied by including different combinations of independent variables. PLS model M2 focused on effect of pretreatment conditions. PLS model M2 provides a reference for evaluating the effect of inclusion of other variable groups into a different PLS model. For example, antisolvents were included in PLS model M3, while type of antisolvent and biomass composition were considered in PLS model M4. PLS model M5 considered all the above-mentioned variable groups with the inclusion of type of HBD in DES and its physicochemical properties.

3.2.1. Analysis of PLS model M2

PLS model M2 was established to determine the effect of six variables related to pretreatment conditions on enzymatic hydrolysis efficacy. Pretreatment duration, temperature, severity factor, solid-to-liquid ratio, particle size, and agitation intensity were set as independent variables, while glucose yield and saccharification duration were set as

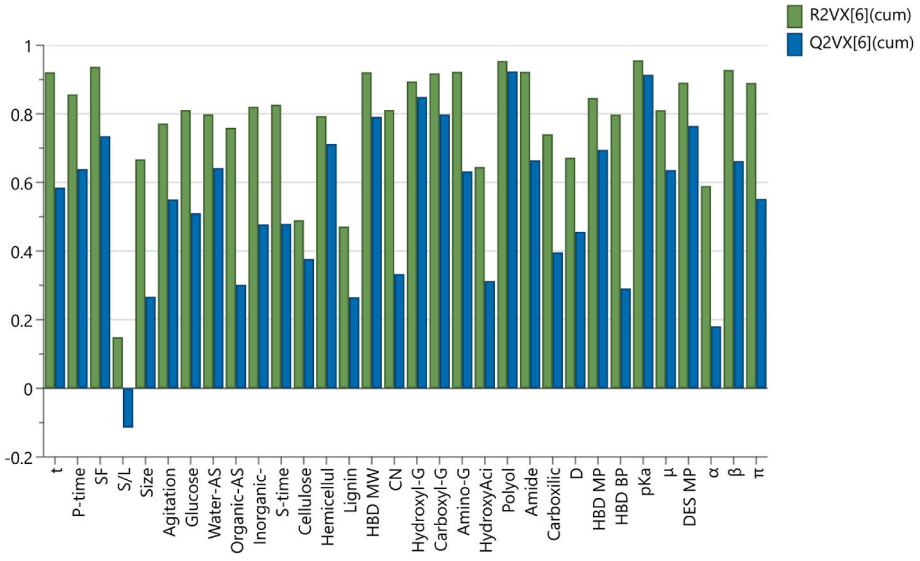


Fig. 2. Variance overview plot for PCA model M1.

dependent variables. In PLS model M2, explained variance for X variables (R2Xcum), explained variance for Y variables (R2Ycum), and predicted variance (Q2cum) were 0.519, 0.703 and 0.677, respectively. Since the model could explain 70.3% of dependent variables variation and had a 67.7% predictive ability, it was considered to have good assessing capability. As shown in Fig. 3(a), the first component (w*c[1]) and the second component (w*c[2]) collectively explained 51.9% variance of the six independent variables. Component 1 was dominated by the severity factor, while component 2 was strongly affected by agitation intensity. The variable importance plot, shown in Fig. 3(b), states that the most significant variables in this model were agitation intensity and severity factor since their values were above 1.

The relationships between the dependent and independent variables were revealed by the coefficient plots in Fig. 3(c) and (d). The size of the coefficient represents the change in the Y-variable when the X-variable varies from 0 to 1. The error bars indicate the confidence intervals of the coefficients. The coefficient is significant (above the noise) when the confidence interval does not include zero. Based on the coefficient plot, glucose yield was positively correlated with pretreatment temperature, severity factor, and negatively correlated to solid loading. On the other hand, saccharification duration had a strong positive correlation with agitation, particle size, and negatively correlated with pretreatment duration. Since nearly all coefficients of variables had high significance and the model had high explained and predicted variance, it can be concluded that pretreatment conditions alone can provide a good explanation of the lignocellulose pretreatment process with DES.

3.2.2. Analysis of PLS model M3

The effect of antisolvent on enzymatic efficacy was studied by including the type of antisolvent in PLS model M3. In the model, pretreatment duration, temperature, severity factor, solid-to-liquid ratio, particle size, agitation intensity, water antisolvent, organic antisolvent, and inorganic antisolvent were set as independent variables, while glucose yield and saccharification duration, were set as dependent variables. In PLS model M3, R2Xcum, R2Ycum and Q2cum were 0.835, 0.756 and 0.721, respectively. Since the model could explain 75.6% of the variation of the dependent variables and had a 72.1% predictive ability, it was considered to have a substantial assessing capability. As shown in Fig. 4(a), the first component and the second component collectively explained 49.2% variance of the 9 independent variables. Similar to PLS model M2, component 1 was dominated by the severity factor, while component 2 was strongly affected by agitation intensity. Apart from agitation intensity and severity factor, inorganic antisolvent and temperature also contribute significantly towards the enzymatic efficacy, as shown in Fig. 4(b).

Even though the inclusion of the variables describing the type of antisolvent helped to increase explained and predicted variance of the model M3 comparing to the model M2, and these variables had a moderate impact on the model according to the VIP chart (only one type of antisolvent had value importance above 1, while the importance of other two types of antisolvents lied between 0.5 and 1), coefficient plots suggest that only a few correlations between enzymatic hydrolysis efficacy parameters and types of antisolvents had sufficient degree of certainty. As shown in Fig. 4(c) and (d), glucose yield was positively correlated with temperature and severity factor, and negatively with water antisolvent. Saccharification duration had a positive correlation with agitation intensity, organic antisolvent, and particle size, and negative correlation with inorganic antisolvent. Although coefficient plots showed that a positive correlation exists between glucose yield with organic and inorganic antisolvents, large error bars crossing the horizontal axis suggested that these correlations were quite uncertain.

3.2.3. Analysis of PLS model M4

In PLS model M4, biomass composition was also considered apart from pretreatment conditions and type of antisolvent. Pretreatment duration, temperature, severity factor, solid-to-liquid ratio, particle size, agitation intensity, cellulose content, hemicellulose content, lignin content, water antisolvent, organic antisolvent, and inorganic antisolvent were set as independent variables, while glucose yield and saccharification duration, were set as dependent variables. In PLS model M4, R2Xcum, R2Ycum and Q2cum were 0.781, 0.803 and 0.776, respectively. The model was considered to have a substantial assessing capability, since it could explain 80.3% of dependent variables variation and had a 77.6% predictive ability. As shown in Fig. 5(a), the first component and the second component collectively explained 49.6% variance of the twelve independent variables. Component 1 was

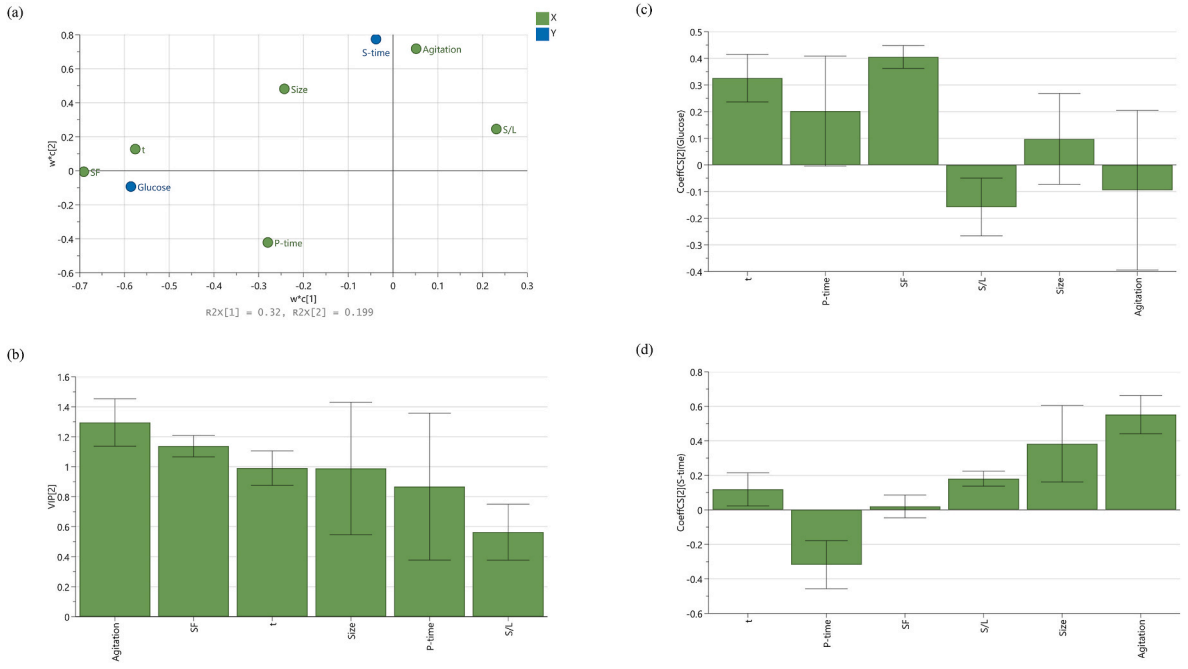


Fig. 3. PLS model M2 for DES pretreatment of lignocellulosic biomass: (a) loading scatter plot; (b) VIP chart; (c) coefficient plot for glucose yield; and (d) coefficient plot for saccharification duration.

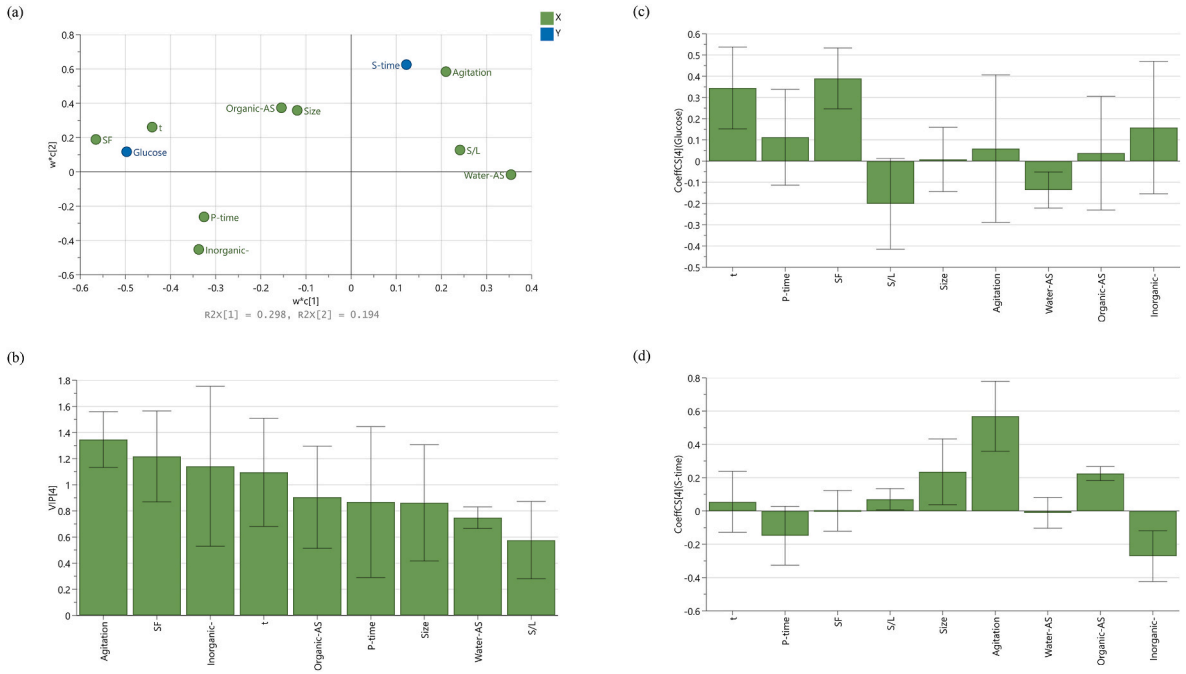


Fig. 4. PLS model M3 for DES pretreatment of lignocellulosic biomass: (a) loading scatter plot; (b) VIP chart; (c) coefficient plot for glucose yield; and (d) coefficient plot for saccharification duration.

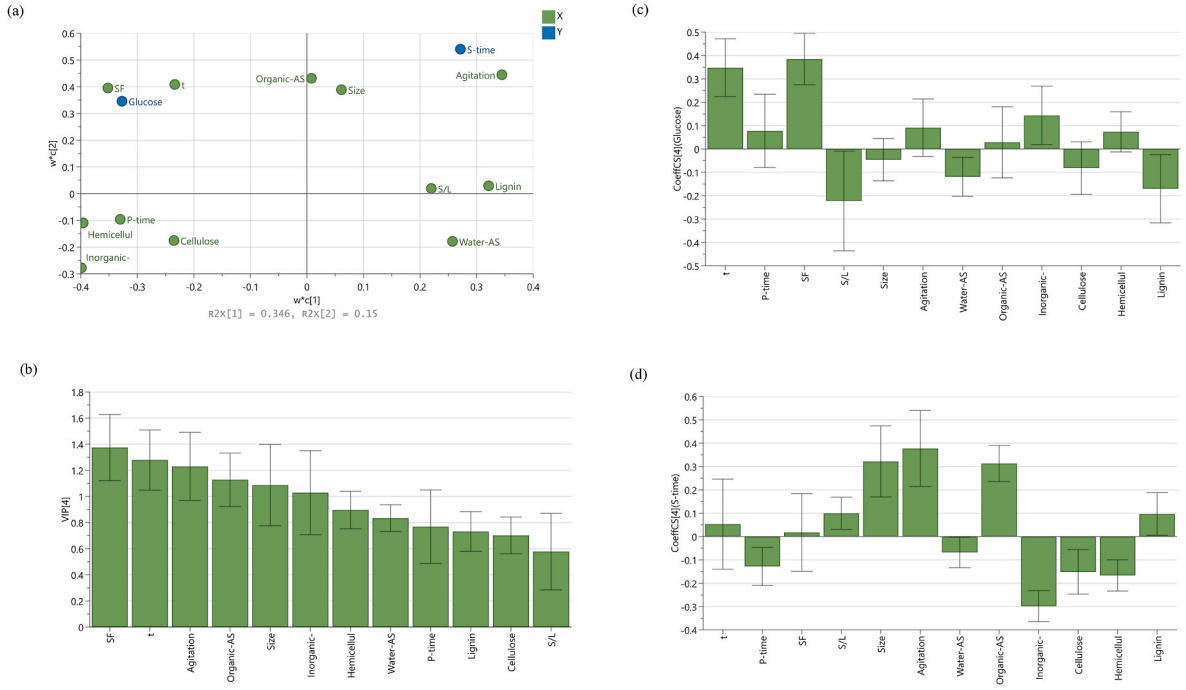


Fig. 5. PLS model M4 for DES pretreatment of lignocellulosic biomass: (a) loading scatter plot; (b) VIP chart; (c) coefficient plot for glucose yield; and (d) coefficient plot for saccharification duration.

dominated by the hemicellulose content and inorganic antisolvent, while component 2 was strongly affected by severity factor, temperature, agitation intensity, particle size, and organic antisolvent. Despite that PLS model M4 was used to investigate the effect of biomass composition, none of them appear in the variable importance plot, as shown in Fig. 5(b). The most significant variables in this model were severity factor, temperature, agitation, organic antisolvent, particle size, and inorganic antisolvent.

The relationships between the dependent and independent variables were revealed in the coefficient plots shown in Fig. 5(c) and (d). Based on these coefficient plots, glucose yield was positively correlated with temperature, severity factor, and inorganic antisolvent, and negatively correlated with solid-to-liquid ratio, water antisolvent, and lignin content. Correlations of glucose with cellulose and hemicellulose contents were uncertain due to large error bars crossing the horizontal axis. Saccharification duration had a positive correlation with solid-to-liquid ratio, particle size, agitation intensity, organic antisolvent, and lignin content, and a negative correlation with pretreatment duration,

inorganic antisolvent, cellulose, and hemicellulose contents.

3.2.4. Analysis of PLS model M5

PLS model M5 was established to study the effect of all groups of variables and on enzymatic hydrolysis efficacy. All dependent and independent variables presented in the Table 2 were used in the model M5. In PLS model M5, R2Xcum, R2Ycum, and Q2cum were 0.488, 0.762, and 0.703, respectively. The model could explain 76.2% of dependent variables variation and had predictive ability of 70.3%. Both the values indicated the good assessing capability of the model. As shown in Fig. 6(a), the first component and the second component collectively explained a 40.2% variance of the thirty independent variables. Component 1 was strongly affected by the inorganic antisolvent, carboxyl group number, hemicellulose content, agitation, and Kamlet-Taft hydrogen bond acidity, while component 2 was strongly affected by severity factor, temperature, organic antisolvent, and particle size. The variable importance plot, shown in Fig. 6(b), states that the most significant variables in this model were severity factor, temperature, particle size, organic antisolvent, agitation intensity, inorganic antisolvent, Kamlet-Taft hydrogen bond acidity, carboxyl group number, and water antisolvent.

The relationships between the dependent and independent variables were revealed in the coefficient plots in Fig. 6(c) and (d). Based on these coefficient plots, glucose yield was positively correlated with temperature and severity factor, and negatively correlated with solid-to-liquid ratio, water antisolvent, lignin content. Saccharification duration had a positive correlation with solid-to-liquid ratio, particle size, agitation intensity, organic antisolvent, polyol HBD, pKa, and Kamlet-Taft hydrogen bond acidity, and a negative correlation with pretreatment duration, water antisolvent, inorganic antisolvent, cellulose content, hemicellulose content, number of carboxyl group, hydroxy acid HBD. Correlations with all other unmentioned variables were considered insignificant due to large uncertainty. Even though the PLS model M5 was considered to have good assessing and prediction ability, its values of cumulative explained and predicted variance were lower than those of PLS models M4. In PLS model M5, physiochemical properties and characteristics of HBD were less significant to glucose yield than other pretreatment variables. Saccharification duration was mostly affected by agitation, composition, and particle size of biomass.

3.2.5. Comparison of PLS models

As can be seen from Table 4, all PLS models for DES pretreatment had rather high cumulative explained and predicted variances, therefore all of these models were considered to have good predictive power and prediction relevance. PLS model M2 assessed the effect of pretreatment conditions. It appeared to be the most inaccurate model with the lowest values of all three variances. Nevertheless, since all three variances had values above 0.5, this model was able to provide satisfactory explanation of DES pretreatment process.

PLS model M3 considered the effect of pretreatment conditions and the type of antisolvent during lignocellulose regeneration on enzymatic saccharification efficacy evaluation parameters. Since both R2Ycum and R2Ycum for this model exceeded 0.75, while Q2cum was above 0.5, the model M3 was said to have substantial prediction power and good prediction relevance. By comparing model M3 against M2, type of antisolvent was an important parameter having significant influence on enzymatic saccharification efficacy evaluation parameters. PLS model M4 included all variables from M3 with the incorporation of lignocellulose compositions as independent variables. Inclusion of lignocellulose compositions increased cumulative explained variance for dependent variables and cumulative predicted variance, however, caused reduction of cumulative explained variance for independent variables.

Table 4
Explained and predicted variance for each PLS model for DES pretreatment.

Model	Number of principal components	R2Xcum	R2Ycum	Q2cum
PLS model M2	2	0.519	0.703	0.677
PLS model M3	4	0.835	0.756	0.721
PLS model M4	4	0.781	0.803	0.776
PLS model M5	3	0.488	0.762	0.703

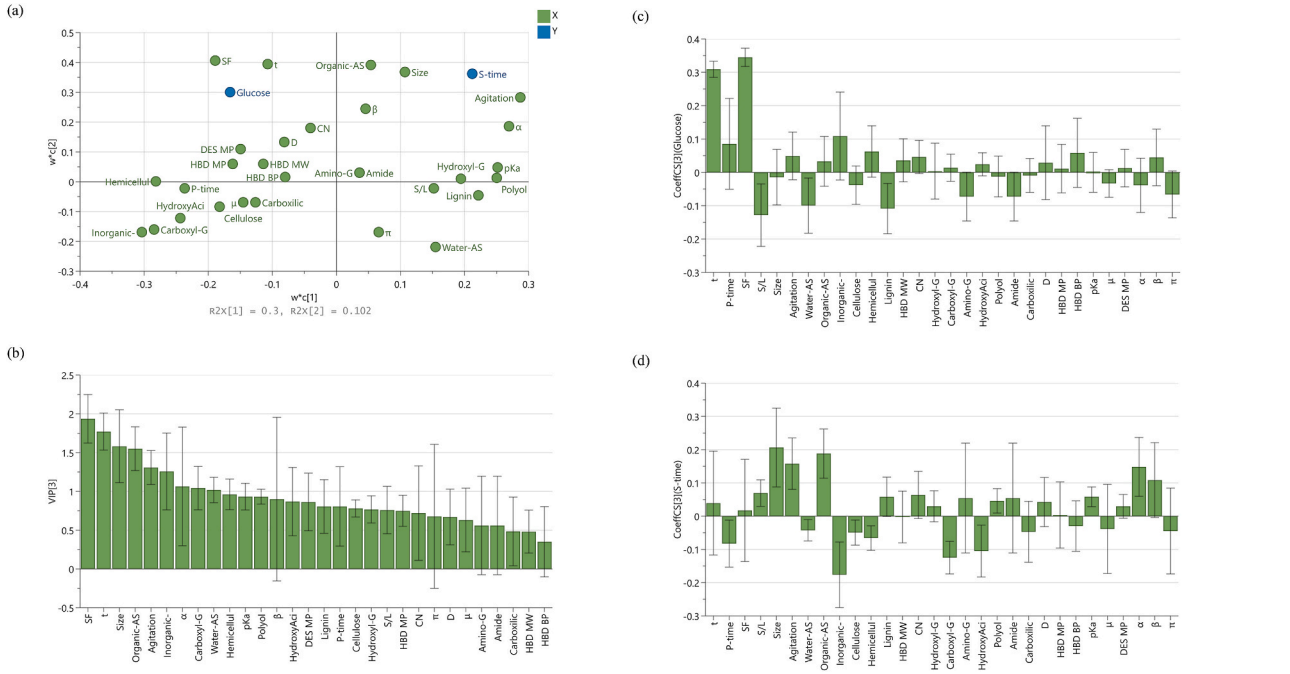


Fig. 6. PLS model M5 for DES pretreatment of lignocellulosic biomass: (a) loading scatter plot; (b) VIP chart; (c) coefficient plot for glucose yield; and (d) coefficient plot for saccharification duration.

Nevertheless, all three of them had values larger than 0.75, hence model M4 was said to have substantial predictive power and prediction relevance. PLS model M5 included all groups of variables. This model had the lowest cumulative explained variance for independent variables and average values of cumulative explained variance of dependent variables and cumulative predicted variance. Overall, model M4 had substantial predictive power for dependent variable and good relevance of prediction.

4. Conclusion

With both PCA and PLS multivariate analysis methods, the relationship between the pretreatment condition, type of antisolvent, biomass composition, type of DES and its physicochemical properties on glucose yield and saccharification duration were analyzed. PCA model for DES pretreatment included six principal components that accounted for 78.8% of the total variability and had a cumulative predicted variance of 0.524, which indicated its good predictive power and relevance. This model suggested that glucose yield could be improved by longer pretreatment duration at a lower temperature but with a large resulting severity factor. The model showed that organic and inorganic antisolvents are more beneficial for glucose yield rather than water antisolvent. The use of carboxylic acid and hydroxy acid HBD was associated with higher glucose yield and shorter duration of saccharification. Glucose yield in this model was favored by larger size particles. Unexpectedly, intensive agitation tended to increase the time required to complete enzymatic saccharification. This could be potentially explained by the non-Newtonian properties of DES, and the formation of shear-thickening fluid upon mixing DES with fine lignocellulose particles. PLS model M4 was selected as the best model to describe the effect of various DES pretreatment parameters on saccharification efficacy evaluation parameters. R2Xcum, R2Ycum, and Q2cum of the model were 0.781, 0.803, and 0.776, indicating its substantial prediction power for dependent variables and prediction relevance. The most significant variables affecting the pretreatment process were severity factor, temperature, type of antisolvent, particle size and agitation intensity. The correlations obtained in this work could play a significant role in guiding the researchers in selecting appropriate pretreatment variables in optimizing the DES pretreatment.

Funding

This work was supported by the UCSI University Research Excellence & Innovation Grant (Grant number: REIG-FETBE-2020/038).

CRediT authorship contribution statement

Salim Massayev: Methodology, Formal analysis, Investigation, Writing – original draft. **Kiat Moon Lee:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank OPTIMUM LLP for sponsoring the SIMCA software.

Appendix A

Table A.1
List of variables for DES pretreatment of lignocellulosic biomass

Full name of variable	Variable type	Dummy variable	Abbreviation in model
Temperature, °C	Numeric		t
Pretreatment duration, h	Numeric		P-time
Severity factor	Numeric		SF
Solid-to-liquid ratio, w/w	Numeric		S/L
Average particle size, micron	Numeric		Size
Agitation intensity, rpm	Numeric		Agitation
Glucose yield, %	Numeric		Glucose
Antisolvent	Categorical	Water, Organic solvent, Inorganic solvent	Water-AS, Organic-AS, Inorganic-AS
Saccharification duration, h	Numeric		S-time
Cellulose content in untreated biomass	Numeric		Cellulose
Hemicellulose content in untreated biomass	Numeric		Hemicellulose
Lignin content in untreated biomass	Numeric		Lignin
HBD molecular weight	Numeric		HBD MW
HBD carbon number	Numeric		CN
HBD hydroxyl groups number	Numeric		Hydroxyl-G
HBD carboxyl groups number	Numeric		Carboxyl-G
HBD amino groups number	Numeric		Amino-G
HBD type	Categorical	Hydroxy acid, Polyol, Amide, Carboxilic acid	HydroxyAcid, Polyol, Amide, CarboxilicAcid
HBD dipole moment	Numeric		D
HBD melting point, °C	Numeric		HBD MP
HBD boiling point, °C	Numeric		HBD BP
HBD pKa	Numeric		pKa
DES viscosity at 25 °C	Numeric		μ
DES Melting point, °C	Numeric		DES MP
DES hydrogen bond acidity in Kamlet-Taft parameters	Numeric		α
DES hydrogen bond basicity in KamletTaft parameters	Numeric		β
DES dipole polarization	Numeric		π

References

Abbott, A.P., Boothby, D., Capper, G., Davies, D.L., Rasheed, R.K., 2004. Deep eutectic solvents formed between choline chloride and carboxylic acids: versatile alternatives to ionic liquids. J. Am. Chem. Soc. 126, 9142–9147. https://doi.org/10.1021/ja048266j.

Al Ameri, M., 2017. Deep Eutectic Solvent Pretreatment for Enhancing Biochemical Conversion of Switchgrass. University of Missouri-Columbia.

Brown, E., Jaeger, H.M., 2014. Shear thickening in concentrated suspensions: phenomenology, mechanisms and relations to jamming. Reports Prog. Phys. 77 https://doi.org/10.1088/0034-4885/77/4/046602.

Degam, G., 2017. Deep Eutectic Solvents Synthesis, Characterization and Applications in Pretreatment of Lignocellulosic Biomass. Theses Diss.

Doherty, T.V., Mora-Pale, M., Foley, S.E., Linhardt, R.J., Dordick, J.S., 2010. Ionic liquid solvent properties as predictors of lignocellulose pretreatment efficacy. Green Chem 12. https://doi.org/10.1039/c0gc00206b, 1967–1975.

Elhamarnah, Y.A., Nasser, M., Qiblawey, H., Benamor, A., Atilhan, M., Aparicio, S., 2019. A comprehensive review on the rheological behavior of imidazolium based ionic liquids and natural deep eutectic solvents. J. Mol. Liq. 277, 932–958. https://doi.org/10.1016/j.molliq.2019.01.002.

Eriksson, L., Byrne, T., Johansson, E., Trygg, J., Vikström, C., 2013. Multi- and Megavariate Data Analysis: Basic Principles and Applications, third ed. Umetrics Academy.

Florindo, C., McIntosh, A.J.S., Welton, T., Branco, L.C., Marrucho, I.M., 2018. A closer look into deep eutectic solvents: exploring intermolecular interactions using solvatochromic probes. Phys. Chem. Chem. Phys. 20, 206–213. https://doi.org/10.1039/c7cp06471c.

Hajilary, N., Shahi, A., Rezakazemi, M., 2018. Evaluation of socio-economic factors on CO2 emissions in Iran: factorial design and multivariable methods. J. Clean. Prod. 189, 108–115. https://doi.org/10.1016/j.jclepro.2018.04.067.

Lauberte, L., Telysheva, G., Cravotto, G., Andersone, A., Janceva, S., Dizhbite, T., Arshanitsa, A., Jurkjane, V., Vevere, L., Grillo, G., Gaudino, E.C., Tabasso, S., 2021. Lignin – derived antioxidants as value-added products obtained under cavitation treatments of the wheat straw processing for sugar production. J. Clean. Prod. 303 https://doi.org/10.1016/j.jclepro.2021.126369.

Lee, K.M., Hong, J.Y., Tey, W.Y., 2021. Combination of ultrasonication and deep eutectic solvent in pretreatment of lignocellulosic biomass for enhanced enzymatic saccharification. Cellulose 28, 1513–1526. https://doi.org/10.1007/s10570-020-03598-5.

Lee, K.M., Ngoh, G.C., Chua, A.S.M., 2015. Ionic liquid-mediated solid acid saccharification of sago waste: kinetic, ionic liquid recovery and solid acid catalyst reusability study. Ind. Crops Prod. 77, 415–423. https://doi.org/10.1016/j.indcrop.2015.09.016.

Lee, K.M., Zanil, M.F., Chan, K.K., Chin, Z.P., Liu, Y.C., Lim, S., 2020. Synergistic ultrasound-assisted organosolv pretreatment of oil palm empty fruit bunches for enhanced enzymatic saccharification: an optimization study using artificial neural networks. Biomass and Bioenergy 139. https://doi.org/10.1016/j.biombioe.2020.105621.

Li, B., Ding, L., Xu, H., Mu, X., Wang, H., 2017. Multivariate data analysis applied in alkali-based pretreatment of corn stover. Resour. Conserv. Recycl. 122, 307–318. https://doi.org/10.1016/j.resconrec.2016.12.007.

Liu, Y., Zheng, J., Xiao, J., He, X., Zhang, K., Yuan, S., Peng, Z., Chen, Z., Lin, X., 2019. Enhanced enzymatic hydrolysis and lignin extraction of wheat straw by triethylbenzyl ammonium chloride/lactic acid-based deep eutectic solvent pretreatment. ACS Omega 4, 19829–19839. https://doi.org/10.1021/acsomega.9b02709.

Liu, Z.H., Qin, L., Pang, F., Jin, M.J., Li, B.Z., Kang, Y., Dale, B.E., Yuan, Y.J., 2013. Effects of biomass particle size on steam explosion pretreatment performance for improving the enzyme digestibility of corn stover. Ind. Crops Prod. 44, 176–184. https://doi.org/10.1016/j.indcrop.2012.11.009.

Lloyd, T.A., Wyman, C.E., 2005. Combined sugar yields for dilute sulfuric acid pretreatment of corn stover followed by enzymatic hydrolysis of the remaining solids. Bioresour. Technol. 96, 1967–1977. https://doi.org/10.1016/j.biortech.2005.01.011.

Lobato-Peralta, D.R., Duque-Brito, E., Villafán-Vidales, H.I., Longoria, A., Sebastian, P.J., Cuentas-Gallegos, A.K., Arancibia-Bulnes, C.A., Okoye, P.U., 2021. A review on trends in lignin extraction and valorization of lignocellulosic biomass for energy applications. J. Clean. Prod. 293 https://doi.org/10.1016/j.jclepro.2021.126123.

Loow, Y.-L., New, E.K., Yang, G.H., Ang, L.Y., Foo, L.Y.W., Wu, T.Y., 2017. Potential use of deep eutectic solvents to facilitate lignocellulosic biomass utilization and conversion. Cellulose 24, 3591–3618. https://doi.org/10.1007/s10570-017-1358-y.

Mahmood, H., Moniruzzaman, M., Yusup, S., Akil, H.M., 2016. Pretreatment of oil palm biomass with ionic liquids: a new approach for fabrication of green composite board. J. Clean. Prod. 126, 677–685. https://doi.org/10.1016/j.jclepro.2016.02.138.

Mahmoudi, R., Emrouznejad, A., Khosroshahi, H., Khashei, M., Rajabi, P., 2019. Performance evaluation of thermal power plants considering CO2 emission: a multistage PCA, clustering, game theory and data envelopment analysis. J. Clean. Prod. 223, 641–650. https://doi.org/10.1016/j.jclepro.2019.03.047.

Nguyen, Q.A., Tucker, M.P., Keller, F.A., Eddy, F.P., 2000. Two-stage dilute-acid pretreatment of softwoods. Appl. Biochem. Biotechnol. 84, 561–576. https://doi.org/10.1385/ABAB:84-86:1–9:561.

Procentese, A., Raganati, F., Olivieri, G., Russo, M.E., Rehmann, L., Marzocchella, A., 2018. Deep eutectic solvents pretreatment of agro-industrial food waste. Biotechnol. Biofuels 11. https://doi.org/10.1186/s13068-018-1034-y.

Ren, H., Chen, C., Wang, Q., Zhao, D., Guo, S., 2016. The properties of choline chloride-based deep eutectic solvents and their performance in the dissolution of cellulose. BioResources 11, 5435–5451. https://doi.org/10.15376/biores.11.2.5435-5451.

Sai, Y.W., Lee, K.M., 2019. Enhanced cellulase accessibility using acid-based deep eutectic solvent in pretreatment of empty fruit bunches. Cellulose 26, 9517–9528. https://doi.org/10.1007/s10570-019-02770-w.

Scelsi, E., Angelini, A., Pastore, C., 2021. Deep eutectic solvents for the valorisation of lignocellulosic biomasses towards fine chemicals. Biomass 1, 29–59. https://doi.org/10.3390/biomass1010003.

Schell, D.J., Farmer, J., Newman, M., McMillan, J.D., 2003. Dilute-sulfuric acid pretreatment of corn stover in pilot-scale reactor: Investigation of yields, kinetics, and enzymatic digestibilities of solids. Appl. Biochem. Biotechnol. 105, 69–86. https://doi.org/10.1385/ABAB:105:1–3:69.

Semerci, I., Ersan, G., 2021. Hornbeam pretreatment with protic ionic liquids: cation, particle size, biomass loading and recycling effects. Ind. Crops Prod. 159 https://doi.org/10.1016/j.indcrop.2020.113021.

Thi, S., Lee, K.M., 2019. Comparison of deep eutectic solvents (DES) on pretreatment of oil palm empty fruit bunch (OPEFB): cellulose digestibility, structural and morphology changes. Bioresour. Technol. 282, 525–529. https://doi.org/10.1016/j.biortech.2019.03.065.

Wang, B., Shen, P., Zhu, W., Pang, Z., Dong, C., 2020. Effect of ground wood particle size on biomass fractionation using p-toluenesulfonic acid treatment. Cellulose 27, 4043–4052. https://doi.org/10.1007/s10570-020-03037-5.

Xia, Q., Liu, Yongzhuang, Meng, J., Cheng, W., Chen, W., Liu, S., Liu, Yixing, Li, J., Yu, H., 2018. Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass. Green Chem 20, 2711–2721. https://doi.org/10.1039/c8gc00900g.

Xu, F., Sun, J., Wehrs, M., Kim, K.H., Rau, S.S., Chan, A.M., Simmons, B.A., Mukhopadhyay, A., Singh, S., 2018. Biocompatible choline-based deep eutectic solvents enable one-pot production of cellulosic ethanol. ACS Sustain. Chem. Eng. 6, 8914–8919. https://doi.org/10.1021/acssuschemeng.8b01271.

Xu, H., Kong, Y., Peng, J., Song, X., Che, X., Liu, S., Tian, W., 2020. Multivariate analysis of the process of deep eutectic solvent pretreatment of lignocellulosic biomass. Ind. Crops Prod. 150 https://doi.org/10.1016/j.indcrop.2020.112363.

Zdanowicz, M., Wilpiszewska, K., Spychaj, T., 2018. Deep eutectic solvents for polysaccharides processing. A review. Carbohydr. Polym. 200, 361–380. https://doi.org/10.1016/j.carbpol.2018.07.078.